Macroeconomic regime, trade openness, unemployment and inequality. The Argentine Experience*

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Abstract

Argentina is an interesting country to be analyzed from the point of view of the evolution of their population's welfare during the last three decades. Being a country characterized in the region by a low level of inequality and reduced poverty incidence, the country has experienced a strong worsening in labour and social conditions, in part as a consequence of the macroeconomic crisis. However, these negative trends were also verified in periods of economic growth.

The recent history of Argentina allows establishing two important conclusions. First, the macroeconomic framework is not indifferent regarding its consequences on the social situation. Second, the negative effects that the macroeconomic configuration has on the labour market and the income distribution persist even after the country returns to its growth path. This makes the complete recovery of living condition standards require very important additional efforts through public policy towards the more vulnerable groups, even under macroeconomic regimes more favourable to employment generation.

The aim of this paper is to analyze the Argentine experience focusing on the interactions among macroeconomic regime, labour performance, income distribution and poverty during the Convertibility Plan and the new macroeconomic after the collapse of this currency board regime. A discussion about different theoretical perspectives is presented and diverse labour and social indicators are analyzed.

Key Words: macroeconomic regime, inequality, trade openness, unemployment.
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INTRODUCTION

The distributive analysis of Argentina is relevant for, at least, two reasons. On the one hand, it is among Latin American countries the one that has experienced the most dramatic changes with respect to income inequality and welfare through the last three decades. Argentina changed from being a country with low unemployment and inequality levels and reasonable social integration to experience high social vulnerability, poverty and inequality rates. On the other hand, the experience of this country indicates that it is possible to verify significant growth of the GDP together with a worsening of the labour conditions and income distribution.

The 1990s are a clear example of these dynamics. In a first phase –when the "Convertibility Plan", a currency board regime, was established–, trade opening and exchange rate appreciation generated a process of productive reconversion, characterized by the displacement of local production and the reduction of labour force requirements deriving in an insufficient job generation in a context of strong GDP growth. The combination of very low elasticity of the labour demand together with a significant raise in the labour supply resulted in high and increasing unemployment rates. Later, keeping the currency board regime through successive external crises deepened the high levels of unemployment and the reduction of the average wages in the contractive phases of the cycle.

This economic and labour market performance during that decade contributed to increase the income inequalities and the gap of welfare among rich and poor in Argentina. The distance of household per capita incomes (HPCI) between the 20% richest and 20% poorest population rose from 10 times in 1991 to 20 times at the end of 2001. Poverty headcount ratio, which in 1991 affected to 29% of the population, reached 35% in 2001 and the indigence duplicated.[1]

In this context, Argentina, during 2002, went through an economic and social crisis of unusual magnitude as a consequence of the collapse of the Convertibility at the beginning of that year. The GDP decreased more than 11%, the unemployment climbed to 21,5% and 57,5% of the population lived in households with incomes below the poverty line.[2]

From the second half of 2002, however, some reversion of the labour market indicators trends has operated, as a consequence of the consolidation of the process of economic growth. Nevertheless, its intensity was differential according to the specific analyzed variable: employment has been recovering very quickly since the end of 2002 and poverty has substantially decreased. Other variables have shown a less intense recovery, as is the case of the real incomes and their distribution. Additionally, the job creation has been characterized by being unable to completely solve the problems of precariousness associated with the generation of non-registered wage earners posts.[3] Lastly, the indicators of inequality, poverty and indigence reveal that the country continues going through a vulnerable social situation in spite of the better behaviour of

[1] Data for the "Greater Buenos Aires" (GBA) urban centre (EPH-INDEC). GBA is the major Metropolitan Area of Argentina, with a population of more than 12 million, representing 37% of total urban population and 33% of total population.
[2] Data for all urban centres in which EPH is collected.
[3] Wage earners not registered in the social security system.

the labour market. In the second semester of 2006, 27% of total population was still poor.[4]

The aim of this paper is to study the Argentine experience focusing on the relationship among the macroeconomic regime, the employment generation, the income distribution and the poverty incidence. This analysis is carried out for both macroeconomic regimes: the Convertibility and the new macroeconomic regime in force after the collapse of the currency board. For this, microdata from the official regular household survey – Encuesta Permanente de Hogares (EPH)– are analysed. EPH is carried out by the National Statistical Office (INDEC), which covers urban areas and collects information especially on labour market variables.[5]

The paper has five sections. The first one analyzes the long-term evolution of the most important social indicators in Argentina and compares it with those of some other Latin American countries. The second section presents a brief description of the macroeconomic and social indicators of the nineties in Argentina, whereas in the following section a discussion about factors associated to the worsening of the labour and distributive conditions of that decade is presented. Section 4 analyses the reversion and perdurability of social indicators in the recovery phase after macroeconomic regime change. Section 5 presents the conclusions.

1. Long-term and regional perspective

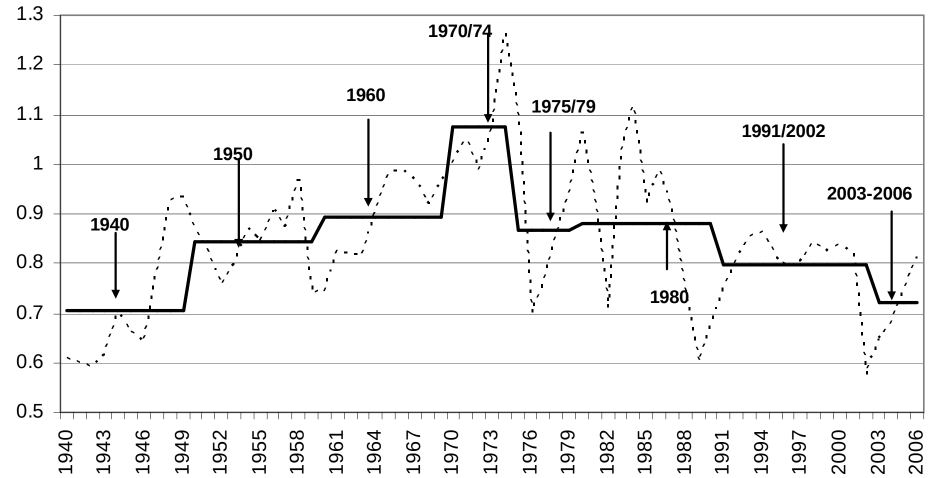
A long-term look allows us to put the more recent developments of the labour market and the inequality in Argentina in perspective. During the last three decades, the country not only experienced macroeconomic instability and deep recession processes but also phases of strong economic growth and relative stability.

However, as shown in Graph 1, beyond these fluctuations, the real wages have been experiencing a downward trend from the maximum obtained in 1974. In particular, the nineties, characterized by very reduced inflation levels and even by deflation episodes, were not able to make the purchasing power of remunerations grow. Instead, it ended up being, on average, lower than that of the previous decade. This systematic worsening makes the wage recovery that started in 2003 insufficient to substantially reverse the previous process. Actually, the average value corresponding to the 2003-2006 period is only higher than the average of the first half of the decade of 1940. Also, it is worth pointing out that over this negative trend four different episodes have reduced even more the average real wages: the freezing of the nominal remunerations imposed by the military government in 1976, the external debt crisis in 1982, the hyperinflation episodes in 1989-1990 and the devaluation in 2002.

[4] Data for all urban centres.

[5] EPH is available since 1974 only for GBA whereas data for all urban centres is available since 1995. Until 2003 it was collected twice a year in 28 urban centres, during May and October. In this year the survey underwent methodological changes (from "EPH Puntual" to "EPH Continua") and produces since then quarterly labour market data and semi-annual poverty data. These methodological changes produced a discontinuity in the labour market series. Due to the lack of necessary information it is not possible to match the series for all variables, difficulting comparison. For a detailed description of the survey's methodology, see www.indec.gov.ar. Finally, unfortunately the analysis ends in the second semester of 2006 due to the lack of microdata onwards.

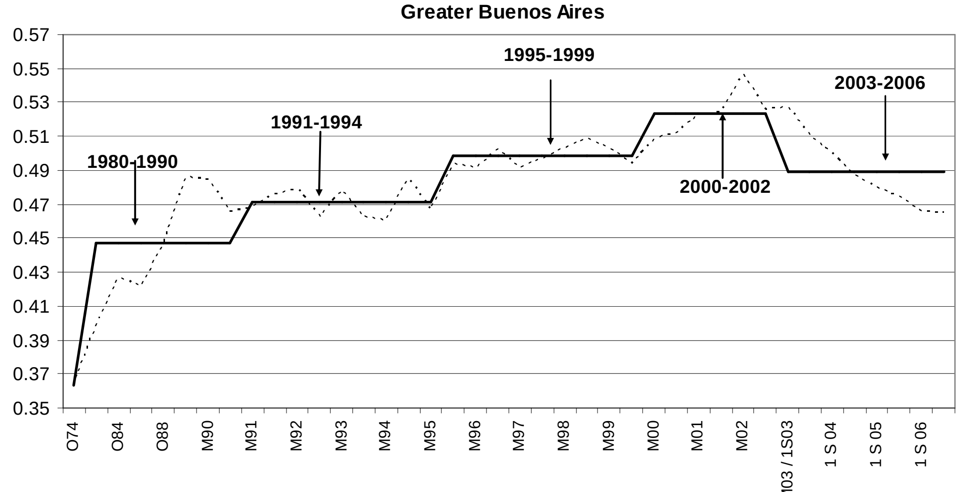
Graph 1
Average Real Wages. Index 1970=1



Source: Author' elaboration based on data from Altimir and Beccaria (1999b).

The reduction of the long-term real wage and the dynamics of the employment were not suffered, however, in a homogeneous way among workers. It resulted in a sustained growth of the inequality levels both of employed population and of households. As shown in Graph 2, Gini coefficient of HPCI has experienced an important growing trend since the middle of the seventies: from 0,363 in 1974 to 0,465 at the end of 2006 in Greater Buenos Aires (GBA). In particular, during most years between the second part of the seventies and 2003 (when the reversion of this trend started) inequality rose and the few years during which income distribution improved, this progress was less intense than the previous worsening.

Graph 2
Gini Coefficient. Household Per Capita Income
Greater Buenos Aires

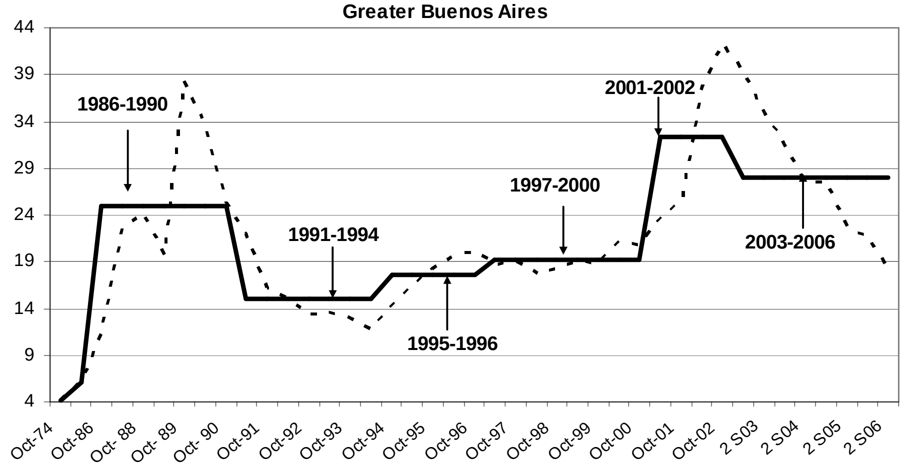


Source: Author' elaboration based on data from EPH (INDEC).

Along the whole period, the factors related to these dynamics have been different. In the second half of the seventies the increase of inequality was fundamentally associated with the raise of inequality among the employed population in a context of strong reduction of real wages in 1976. Following Altimir *et al.* (2002), the higher inequality would be explained by the employers' policy of protecting the most skilled workers from the deterioration of the real remunerations. During the eighties, the labour income gaps among workers remained, even after the return of democracy, in a context of strong macroeconomic instability and stagnation. At this period, furthermore, this behaviour went together with a slight increase of the unemployment rate, fundamentally among the members of the poorest families. This worrying distributional situation was worsened by hyperinflation episodes at the end of the decade. As it will be seen later in detail, the stabilization and economy growth during the nineties did not result in distributive improvements. On the contrary, income inequality remained relatively constant during the first half of this decade experiencing later a sustained growth. Finally, since 2003 a reversion of this trend has operated but inequality remains very high.

The behaviour of the average real wages and the distributional dynamic also had a direct impact on poverty incidence.[6] In particular, the percentage of poor households grew significantly, going from representing 4% of the households in 1974 to 18% in the second semester of 2006 in GBA.

Graph 3
Household Poverty Rate
Greater Buenos Aires



Source: Author' elaboration based on data from EPH (INDEC).

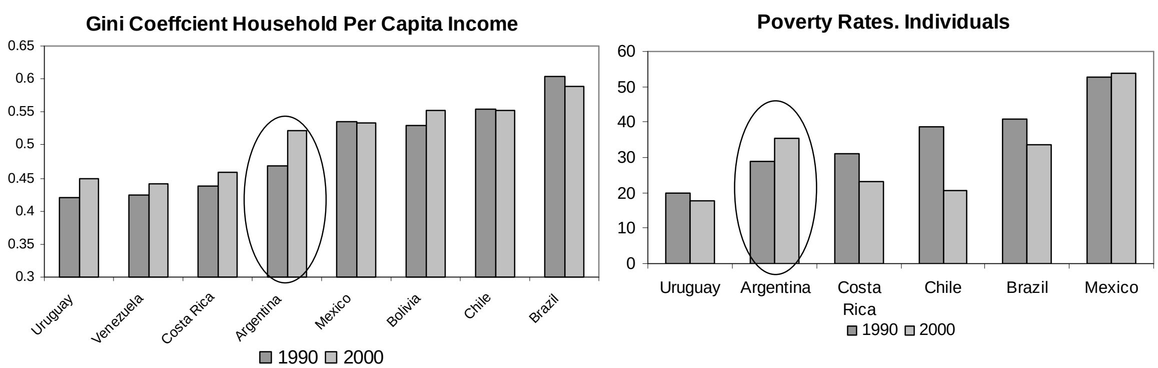
As it is observed in Graph 3, although the poverty reduction in the first half of the nineties was very important, the average incidence in those years went higher than that registered before the hyperinflation episodes of 1989 and 1990. On the other hand, after a new growth period registered between the end of 1994 and the end of 1996, poverty remained relatively constant until 2000 but, again, in levels higher than those of the previous phases. The collapse of the currency board regime and the consequent devaluation made poverty reach levels that had never been registered before in the country. Although since 2003 the upward trend of the poverty rate is reverted, the average of this last period is still one of the highest of the last three decades (with the only exception of the 2001-2002 period).

[6] In Argentina, identification of poor households is done through the absolute poverty line approach. The incidence of poverty is the proportion of households (or population) that cannot afford a basic basket of goods and services. The value of the reference basket is calculated based on domestic prices and patterns of consumption.

It can be observed, then, that the degree of social deprivation that appears after each crisis, although lower than that registered during its development, is significantly higher than the pre-crisis level. Along this process, a certain erosion of the consensus on the satisfactory social conditions is verified and, as a result, situations which were previously considered as intolerable became socially acceptable.

The regional dimension also helps to contextualize the recently mentioned developments. Argentina, together with Costa Rica and Uruguay, has historically been one of the Latin American countries with better social indicators, characterized by a low degree of social polarization and by a large middle class. During the last decade, the experience of the countries of the region has not been homogeneous in terms of income distribution and poverty. In particular, while inequality decreased in Brazil on the contrary, Uruguay, Costa Rica, Venezuela and –especially– Argentina experienced the opposite process (Graph 4). In this sense, a certain reduction has been verified in the gap between countries with respect to the inequality indicators (although it is still very wide) as they rose in the last years in those countries with lower degree of initial inequality, while they reduced in the most inequitable ones.

Graph 4
Inequality and poverty in selected Latin American Countries



Source: Author' elaboration based on BADEINSO (ECLAC).

A similar panorama arises when comparing the evolution of poverty incidence in the case of Argentina with that of the countries of the region. In particular, the strong increase in poverty rates registered during the nineties has been clearly higher than the one registered in most of the countries of the region. In this sense, there is a striking contrast between the Argentine performance and that of its neighbouring countries – Chile, Brazil and Uruguay– since poverty incidence decreased in the first two cases while it remained relatively constant in the third one (Graph 4).

This brief comparative analysis with some of the most important countries in the region (which is by no means exhaustive) indicates that although the distributive changes in Argentina were not opposed to what happened in other Latin American countries, these have been verified more intensely. In fact, during the last thirty years Argentina was one of the Latin American countries that experienced the most important worsening in income distribution and poverty.[7] For this reason, it is interesting to analyze in more

[7] For more details about inequality and polarization in Latin America see, for example, Altimir (1997), Londoño and Székely (2000), Gasparini *et al.* (2006).

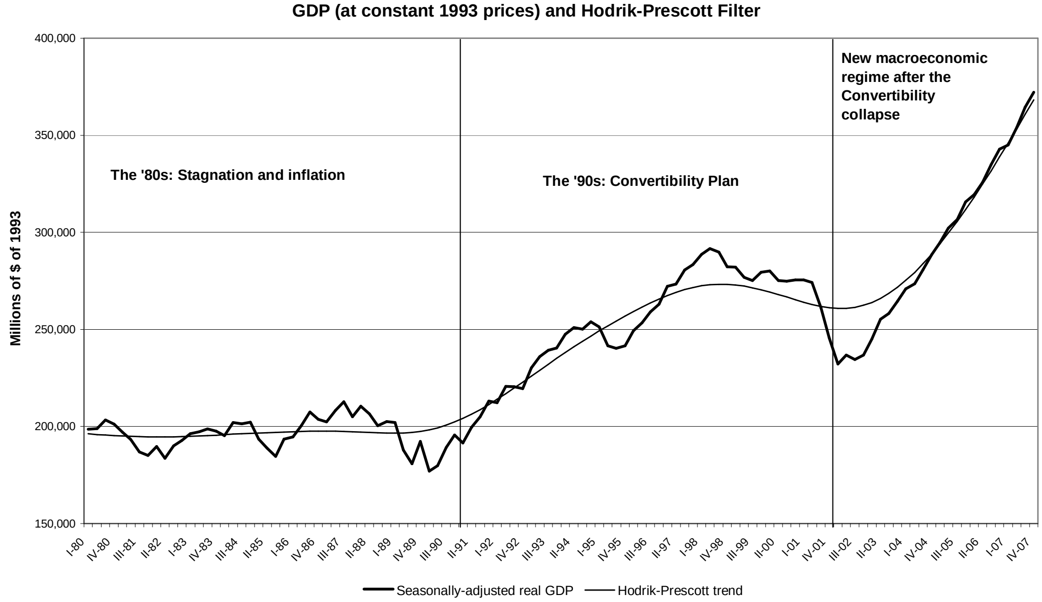
depth the macroeconomic and labour changes associated to the significant changes in inequality and poverty in the country.

2. Macroeconomic regime, labour market and distributional performance during the nineties

After decades of macroeconomic instability, the Convertibility Plan implemented in 1991 was based on the establishment of a fixed exchange rate and the convertibility of the national currency with the dollar at a parity of $1=U$1.[8] Strong and quick commercial and financial liberalization policies were implemented in order to contribute to the convergence of domestic and international inflation. Together with this stabilization plan a vast set of structural reforms was carried out: privatizations of state-owned enterprises, state reform and modification of the pension system were some of the most important. This process of reforms in favour of the market based on the "Washington Consensus" also included intense policies of labour market deregulation and flexibilization.[9]

Initially, these policies contributed to reach a rapid decline of the inflation levels and very high GDP growth rates. However, although the configuration achieved a perdurable success regarding the strong reduction of inflation rates, the same was not verified with respect to economic growth. In particular, during the second part of the decade the GDP experienced strong contractive phases, especially from 1998 to 2002 (Graph 5). Along these years GDP dropped around 20% where 60% of this fall occurred during 2001.

Graph 5
GDP (at constant 1993 prices) and Hodrik-Prescott Filter



Source: Author' elaboration based on data from INDEC.

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[8] Monetary Base must be guaranteed with foreign reserves. For more details about the Convertibility regime, see, for instance, Heymann (2001), Damill *et al* (2002, 2003).
[9] For a discussion about the labour market deregulation, see Beccaria and Galín (2002).

As mentioned in Damill, Frenkel and Maurizio (2007) "*the Argentine macroeconomic experience in the nineties is an example of a more general pattern of external crisis*". First, the capital inflows, as a consequence of the higher domestic interest rate in comparison to the foreign, contribute to reach high GDP growth rates. In this expansionary business cycle, domestic demand increases leading to a rapid raise in the net imports. This behaviour is fostered by the real exchange rate appreciation which contributes to a growing trade worsening.[10] The trade deficit together with the growing deficit in the factor services account lead to a structural and increasing current account deficit. In this context of progressive external vulnerability the country risk premium and the interest rate rise, bringing capital inflows to an end, with the resulting contraction of foreign reserves. The unsustainability of the external debt leads to the financial crisis and the collapse of a regime based on a currency board rule.

Clearly, in this regime domestic demand becomes highly dependent on the balance of payment results: accumulation of international reserves automatically generates an expansion of the monetary base and the credit inducing a growth of the domestic demand and activity. On the contrary, the decline of the reserves in the Central Bank provokes a monetary reduction with the consequent contraction in the domestic activity level. However, even when reserves are growing, increasing external capital inflows are needed in order to maintain a given rate of economic growth.

From the point of view of the employment generation, the new prices configuration that came up under the Convertibility regime had important consequences on the performance of the labour market.[11] Although the high growth rates of the first years of the nineties contributed to the increase of the employment in the non-tradable sectors, the trade opening and the exchange rate appreciation seriously attempted against the employment creation in the industrial sector. At the same time, the reduction of the prices of capital goods in relation to labour resulted in a substitution process of the second for the first across different productive sectors. All that strongly weakened the employment requirement even when the economy exhibited, at the beginning of the nineties, a vigorous growth.

During this whole decade it is possible to identify three stages with clearly differentiated unemployment dynamics, as shown in the following table.

	Growth rates (p.p)			Decomposition of the unemployment variation		Anual elasticity of labour demand (average in each subperiod)
Subperiods	Participation rate	Employment rate	Unemployment rate	Variation of the employment rate	Variation of the participation rate	
M91-M95	5.0	-1.7	13.9	30%	70%	0.157
M95-O98	-0.5	2.8	-6.9	88%	12%	0.617
O98-O01	-1.0	-3.5	5.7	133%	-33%	-0.421
May91-Oct01	3.5	-2.4	12.7	46%	54%	0.114

Source: Author' elaboration based on data from EPH (INDEC) and Ministry of Economy

[10] It is important to remark that the real exchange rate was already appreciated at the beginning of the Convertibility and this appreciated level lasted throughout the nineties.

[11] For an in depth analysis of the labour market during the Convertibility see Lindemboin (1996), Altimir and Beccaria (1999a), Altimir *et al.* (2002), Damill *et al.* (2002), Beccaria and Maurizio (2005).

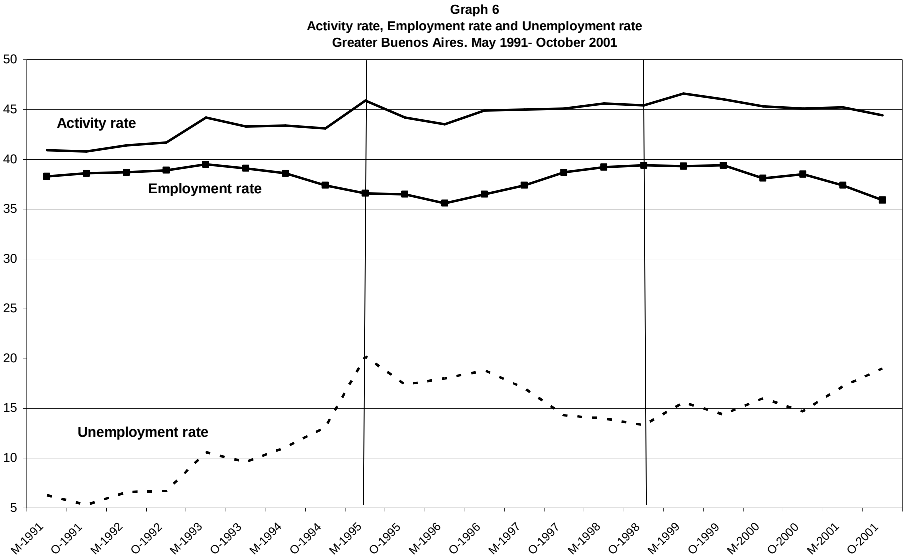
- *May 1991 – May 1995: Macroeconomic growth, low labour demand, increasing labour supply and unemployment.* This subperiod was characterized by high economic growth rates that only resulted in a weak employment creation. The average elasticity of labour demand (employment-product elasticity) was very low, in the order of 0,15. What is more, the employment rate went down since 1993, several months before the macroeconomic crisis of the middle of the nineties (Graph 6). Together with this behaviour, the significant raise in the labour force implied an important and systematic increase of unemployment, which meant that in 1993 it had already reached rates of two digits (Graph 6). In fact, the growing participation rate explained 70% of the increase in the unemployment rate. However, the remaining 30% was due to the reduction in the employment rate.

- *May 1995 – October 1998: Employment growth and unemployment reduction.* After the recession associated to the "tequila crisis", the economy recovered and in this period the employment creation went together with the product growth. In these years the elasticity of labour demand increased (around 0,62) leading to higher generation of new jobs. This behaviour together with certain stability in the participation rate allowed the reduction of unemployment. However, neither employment nor unemployment rates reached the previous levels. In fact, the maximum employment rate during the convertibility was obtained in May 1993 while the minimum unemployment rate was registered in 1991 (Graph 6).

- *October 1998 – October 2001*: *Strong macroeconomic and labour market worsening.* Since the beginning of a new macroeconomic crisis and until the chaotic collapse of the Convertibility, the economic recession generated an additional impulse on the growing trend of unemployment. This worsening was explained completely by the reduction in the employment rate given that the participation rate even decreased. At the end of 2001, the open unemployment rate was near 20% (Graph 6).

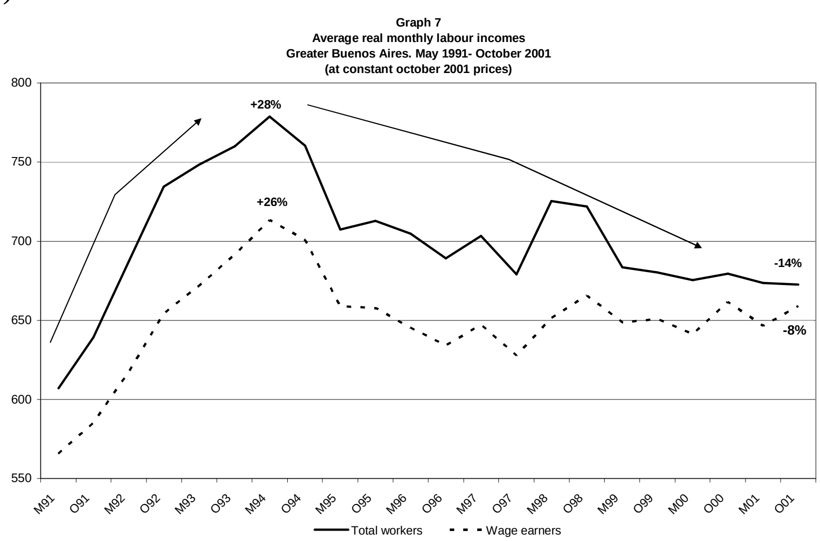
Along the whole period (May 1991 – October 2001) both the increase of the labour supply (3,5 p.p) and the reduction in the labour demand (-2,4 p.p.) explained the strong growth of the unemployment rate (12,7 p.p.). In particular, the first factor accounted for 54% of the total growth in the unemployment rate whereas 46% was explained by the reduction in the employment rate.

Finally, one additional comment should be made with respect to the factors associated to the growth of the activity rate. In particular, it is worth differentiating the "additional worker" from the "encouraged worker". Cerrutti (2000) suggests that the strong increase in female labour force participation in Argentina during the nineties was a response to increasing unemployment and job instability, factor associated to the structural reform since 1991. In particular, women decided to enter the labour force in order to reduce household income decrease and instability. Marshall (2000) mentions that due the reduction of employment and wages of male adults, adult women and youths (both men and women) had to move into the labour force. This was associated, in the case of some young workers, with the abandonment of the educational system.



Source: Author' elaboration based on data from EPH (INDEC).

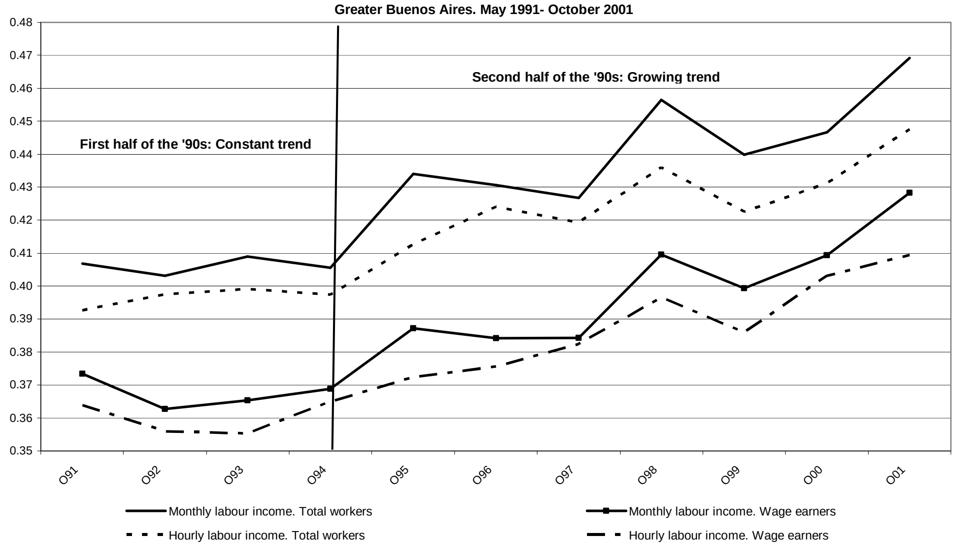
These occupational cycles were verified together with different labour income dynamics (Graph 7). In particular, in the first stage of the Convertibility, the real income achieved a certain recovery as a consequence of the nominal income growth in a context of decreasing levels of domestic inflation. Between the beginning of the nineties and May 1994 average real wages grew 28%, on average, reaching in that moment the maximum of the whole decade. The "Tequila" crisis also had negative effects on the real incomes: between May 1994 and October 1996 the real incomes of the employed population suffered a reduction of 12%. From that date and until October 1998 the incomes slightly recovered, without reaching the maximum registered in the first semester of 1994. Then they experienced a new descending phase until the end of the regime. In October 2001 the average remunerations of the employed population were over the levels of the beginning of the decade in only 11%, which represented a fall of 14% regarding the maximum achieved in 1994. A very similar dynamic was verified among wage earners (Graph 7).



Source: Author' elaboration based on data from EPH (INDEC).

As mentioned, the initial recovery of the remunerations did not allow a reduction of inequality. Between 1991 and 1994 income distribution among the employed population remained relatively constant while the period 1995-2001 showed a process of strong concentration. This growing trend of the Gini coefficient was verified among both the monthly and the hourly labour incomes (for total workers and wage earners) but more intensely in the former than the latter due to the unequal reduction in worked hours, especially among total workers (Graph 8).

Graph 8
Gini Coefficient
Greater Buenos Aires. May 1991- October 2001

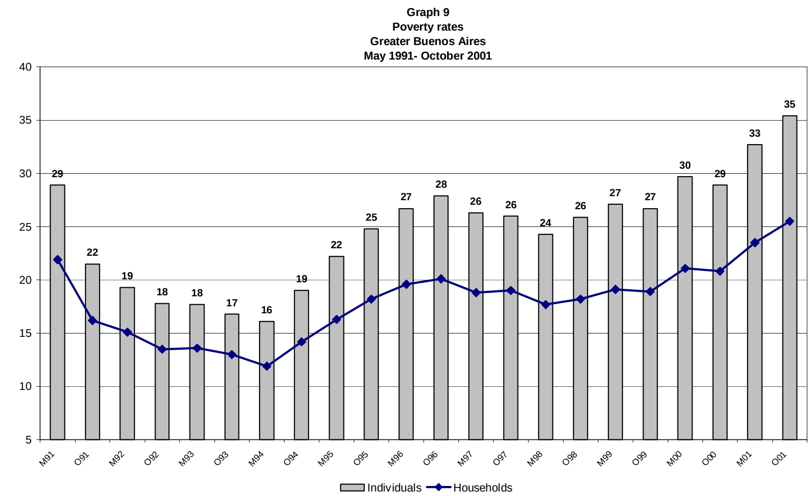


Source: Author' elaboration based on data from EPH (INDEC).

At the same time, this dynamic affected the inequality of the family incomes (Graph 2). While at the beginning of the decade the relationship of the HPCI between the fifth and first quintile was 12 times, it rose to 20 times towards the end of the currency board regime.

The performance of the labour market variables also had a direct impact on poverty levels. Between 1991 and 1994, poverty and indigence decreased as a consequence of the increase in employment and in real remunerations. However, in 1996 this improvement in welfare had already been completely lost. After a slight reduction between 1996 and 1998, these indicators experienced a strong growing trend. Towards October 2001, before the ending of the Convertibility, poverty reached 25% of the households where 35% of the population lived (Graph 9).

Summing up, the behaviour of labour demand under this new regime intensified some characteristics that had begun to emerge in previous years like precariousness, income instability and inequality. Even when the Argentine economy was accumulating a progressive deterioration since the mid-seventies, the Convertibility accelerated this process substantially. Labour difficulties and inequality were exacerbated towards the end of the decade, when the currency board regime became clearly unsustainable. Therefore, the magnitude of this crisis in the months following the devaluation of the Argentine Peso, expressed through the significant contraction of the level of activity, the employment and the income, reflected the important disequilibria that were accumulated during the previous decade.[12]



Source: Author' elaboration based on data from EPH (INDEC).

3. Factors associated to the distributional worsening during the nineties: the role of the macroeconomic regime, the labour market and technological changes

As follows from previous section, the labour market performance was in the centre of the distributional worsening in Argentina, where the growing labour income disparities and unemployment directly impacted on household income inequality.

The increasing wage gap could be associated to the changes in the employment structure as well as the growing returns to individual or job characteristics. Mincer Equations[13] of the hourly labour income carried out exclusively for wage earners shown in Table 1 suggest that along the nineties the changes in the returns to different characteristics have been very significant. In particular, the reduction and relative stability of the inequality among wage earners during the first years of this decade (1991-1993) was result of a reduction of the returns to intermediate educational levels (incomplete and complete high school) together with a certain stability of the returns to university level. Also, a strong decrease of the wage gap between registered and non-registered wage earners was verified.

The significant inequality growth dynamic from 1993 onwards was associated with a process of raise in the educational returns across all levels together with a strong increase of the wage gap between registered and non-registered wage earners. Also, wage differences according to the size of the firm[14] significatively widened.

[12] Beccaria *et al.* (2005).
[13] The equations were corrected by sample selection bias ("lambda" covariate). The control group is conformed by: women with complete primary, non household head, not married, registered wage earner, working full time in manufacture activities and in small enterprises.
[14] Measured by the number of employees working in the firm.

Given the importance of the changes in the educational returns[15], the diverse hypotheses on this distributional behaviour are based on explanations about the wage gap among workers with different skills.

In particular, a very widespread argument about the worsening of the hourly wages' distribution during the nineties in Argentina is based on the so called "Unified Theory"[16] [17] which, starting from a simple competitive model of supply and demand applied to the labour market, shows how an increase in the demand for higher education levels that exceeds the increase in its supply, generates a demand excess in these groups increasing the education returns and worsening, in this way, the labour income distribution. In developed countries this "standard explanation" has frequently been used to explain both the fall in wages of the less educated groups in the US and the increase of unemployment of these groups in European countries.[18] [19] This hypothesis, which tries to explain the shift in the labour demand from unskilled to high skilled workers, in turn, is based on increased openness and integration of the economy[20] and on the technological change biased towards more skilled labour.

In the particular case of Argentina, it is argued that these processes have an impact on the employment structure through two different channels. On the one hand, through reallocations of resources across productive sectors. Stolper-Samuelson theorem is used to indicate that, given that Argentina would be abundant in natural resources and skilled labour force relatively to its main commercial partners in the region[21], trade opening would imply a shift in production and employment towards sectors that use these production factors more intensively, with the consequent increase in their returns.[22] On the other hand, trade liberalization would imply a reduction in the price of external capital goods as well as incorporation of new technology affecting, in both cases, the use of productive factors within sectors. Specifically, since a complementarity between technology and skills is supposed to exist across most productive sectors and

[15] Microeconometric decompositions confirm the importance of the increase in the returns to education in the growing income inequality. See, Gasparini *et al.* (2004), Beccaria and González (2006).
[16] Blank (1994, 1997).
[17] Atkinson (1999) calls this process "Transatlantic Consensus".
[18] Howell and Huebler (2001) find little evidence of a trade-off between inequality and unemployment across OECD countries. As they mentioned *"A convincing explanation of differences in earnings and employment trends across developed countries requires moving beyond simple supply and demand stories"*. They suggest a "neoinstitucionalist" vision of the labour market where *"bargaining power and labor market institutions can be expected to play important and complicated roles, with outcomes not always consistent with the determinate textbook model"*.
[19] Atkinson (1999) also has a critical review of the "Transatlantic Consensus". In particular, he critiques the extended belief that the raising inequality has been inevitable due to this process, as a result of the technological change or the trade globalization. In his vision, changes in the social norms have had an important role where a shift from a redistributive pay norm to a dominance of the market force had been verified with the consequence of widening wage gaps.
[20] In particular, this vision points out the effects of the trade opening of the US and European countries with the NICs (Atkinson, 1999).
[21] There is not complete agreement about the relative abundance of productive factors in Argentina. Galiani and Sanguinetti (2003) suggest that Argentina is abundant in unskilled labour in comparison to developed countries like United States and European Union's countries. However, relative to Latin American countries and China, Argentina is abundant in skilled labour, as argued by Porto and Galiani (2008). Keifman (2006) argues that the country has comparative advantage in natural resources, mainly, in land.
[22] One critical view about theoretical foundations of the neoliberal propositions regarding the economic openness and income inequality is found in Keifman (2006). In particular, Keifman founds that these propositions are not rigorously grounded on mainstream economics.

occupations, the process of technological improvement and capital incorporation also generates a demand towards higher levels of skills.

In comparison with other countries in the region, this argument suggests that Argentina had a higher increase in inequality as a consequence, partly, of the faster speed at which the reforms were carried out, which would probably have had a higher impact in terms of factors and productive sectors reallocations. Also, given that Argentina is technologically more developed than other Latin American countries, the aforementioned changes would have had a deep impact on the wage structure. Finally, the higher educational level would have facilitated a wide diffusion of these technological changes.

However, this perspective does not incorporate the particular local conditions on which the important transformations of the nineties operated. Specifically, this vision does not take into account the role of the Argentine macroeconomic regime during the nineties and its impacts on the aggregate dynamism of the labour market emphasizing only on labour supply conditions and the disequilibria between these and a labour demand biased towards higher skills.

The perspective adopted in this paper on the functioning of the labour market suggests that the macroeconomic regime determines the global performance of the labour market and it has, through this channel, direct impact on the level and distribution of welfare. In particular, in the process of the increasing labour incomes inequality experienced during the last decade in Argentina, an important explaining factor was the low dynamism of the aggregate employment demand and of the persistently high level of unemployment, both determined, at least in part, by the economic configuration during the Convertibility regime.[23] As was already mentioned, the real exchange appreciation, together with very fast trade openness, negatively affected the structure of domestic production already from the beginning of this regime, implying that the employment generation stagnated before the first contractive phase. The inflexibility of the currency board regime implied that different exogenous shocks were transmitted directly to the labour market.

The existence of a very high global labour force surplus not only affects household's income inequality, given that its incidence is higher among members with lower educational levels. On the contrary, the high unemployment can also have important negative impacts on the average remunerations of workers, as shown by the "wage curve".[24] Also, since the "wage curve" can be different for each group of employed people, unemployment could imply a higher inequality among workers.[25] Therefore, the increase in the unemployment rate affects less educated people more intensely both, directly, because of its higher relative incidence and, indirectly, because of its higher negative impact on their wages. Moreover, contexts of very high labour force surplus favour the workers acceptance of more flexible labour conditions as well as generating a

[23] The privatization process also contributed to the destruction of job posts but explains only one small part of the increase of the opening unemployment (Altimir *et al.*, 2002).
[24] The "wage curve" links the wages with the unemployment rate. Specifically, this curve refers to the inverse relationship between a worker's wages and the unemployment rate. Blanchflower and Oswald (1994, 2005) find that US have a wage curve and that its elasticity is approximately –0.1. Also, Howell and Huebler (2001) suggest that higher levels of unemployment implied lower wages and predict a positive association between earnings inequality and unemployment.
[25] For an estimation of the Argentine case, see Damill *et al.* (2002).

"competition for job positions" where the most educated workers displace those of lower skills from their jobs (generating a process of overeducation among skilled workers).[26]

A central theoretical discrepancy between both views refers to the distributive impact of unemployment. Whereas for the first one unemployment does not impact on the distribution of labour income[27], in the second one the *generalized* unemployment not only limits the increase in aggregate real wages, but also induces a differentiation process among wages of different groups of workers, according to the degree in which each of them are exposed to the competition for their job positions. This argument is valid regardless the factor associated to the increase of the unemployment (i.e. workers losing their jobs or people entering the labour force from inactivity). In turn, different theoretical perspectives (as efficiency wages theory or labour markets segmentation theory) can contribute to this vision.

Of course, in our argumentation the distributive impact of the technological change is not ignored. However, our vision puts emphasis on the role of the aggregate labour demand and not only on the employment structure and the adjustments between labour supply and labour demand.

Additionally to theoretical aspects, another relevant issue refers to existing empirical evidence for Argentina supporting each of these perspectives. On the one hand, the first of these arguments is fundamentally based on the idea that, given that during the nineties an increase in the educational level of the labour force was verified together with an increase in educational returns, these facts are *interpreted* as evidence of a shift in the composition of labour demand biased towards skilled workers. That is, in the case of Argentina, there is no direct evidence of changes in the composition of the labour demand but rather they are inferred from the behaviour of labour supply and of the wage premium for skilled workers. However, this configuration of relative wages can also be consistent with other argumentations, especially in the context of high persistent unemployment. Maurizio (2001) found that the educational level of the wage earners increased in all occupations during the nineties, a result that sheds doubts about this indirect evidence. If the educational level of workers could be interpreted as an indicator of technological improvement, one would expect that the increase in skills was differentiated according the exposition of the occupation to technological changes. On the contrary, this evidence seems to be more related to the increasing skilled labour supply in a context of global labour surplus.

On the other hand, the existing evidence does not seem to completely support this type of argument. In the first place, the increase of inequality among workers was verified in the second half of the nineties whereas in the first half they stayed relatively constant, as shown in Graph 8. Nevertheless, as was mentioned before, the integration process was carried out in the first years of this decade, indicating a timing unbalance between both

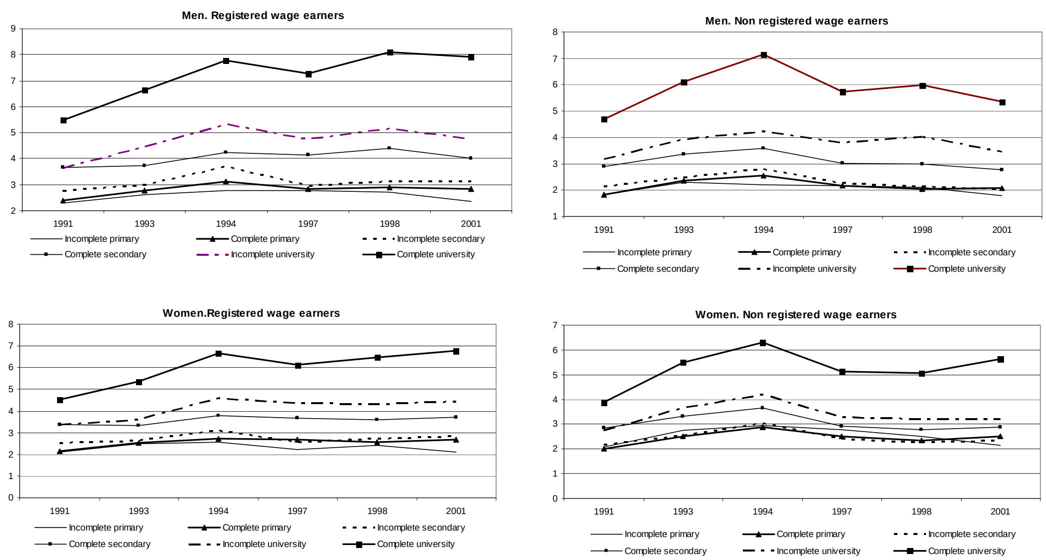
[26] For the Argentine case, see, for instance, Maurizio (2001), Groisman (2003).

[27] The argument is based on in the idea that unemployment growth was caused by an increase of the labour force instead of a reduction of employment. Following this reasoning, suppose one individual who was previously inactive, enters the labour force and becomes unemployed. This individual will not obtain any labour income in either case, so that his entering the labour force does not imply any changes in the wages' distribution. However, it is worth mentioning that, first, unemployment was not only associated to an increase of the labour force (as shown above); second, even if this was the case, the pressure on wages of the unemployed individuals is not the same as that of people who are not part of the labour force.

processes. It could be argued, however, that there is a certain lag between the openness and the technological changes, and its distributive impact. However, in this case, it would be necessary, at least, to keep in mind that the second half of the decade was characterized by strong macroeconomic disequilibria and by a deep recession from 1998 to 2002, which makes it difficult to argue that distributive worsening is *only* a consequence of technological changes.

Secondly, the hypothesis of excess in demand for skilled labour should not only imply relative increases in wages but absolute increases in remunerations as well as a non-increasing unemployment rate for these groups of workers. However, in the case of Argentina, the increments in educational returns in the second part of the nineties were verified in a context of fall in the real and even nominal incomes, as shown in Graph 10.

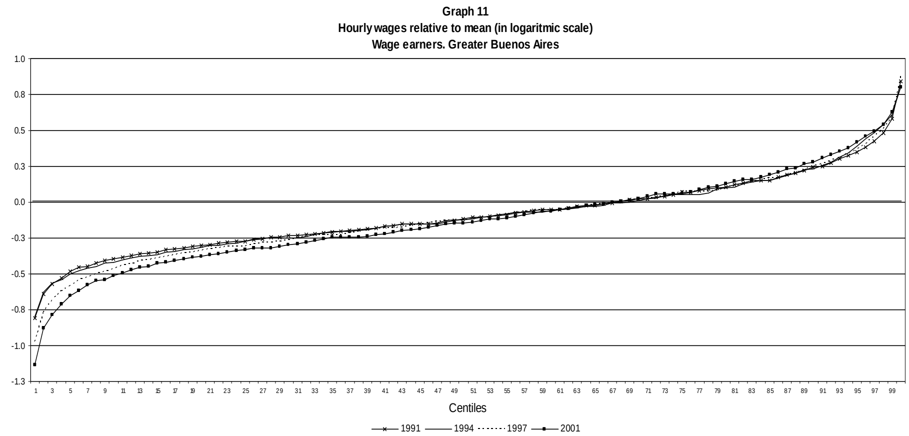
Graph 10
Estimated real hourly wages
Greater Buenos Aires



Source: Author' elaboration based on data from EPH (INDEC).

In particular, nominal and real wages of less educated workers increased up to 1994 and then systematically decreased during the second part of the nineties, especially among of non-registered wage earners. In the case of workers with university level, their wages also grew in the first half and then remained relatively constant until 2001 in the case of registered wage earners. However, non-registered wage earners with university level also experienced a reduction in their labour income from 1994 to 2001. The disparities between wages of registered and unregistered workers also suggest that not only the educational level affects the wage gap but also the type of job is a very relevant factor associated to wage inequality.

Therefore, the higher wages' inequality seems to be due to, at least in this subperiod, the deeper reduction of income among non skilled workers and not because of a higher increase of wages of most educated workers. The disarticulation of certain labour market institutions which tend to protect particularly the lowest wages (i.e. minimum wages, trade unions, collective bargaining) verified in the nineties, would also have contributed to these trends.[28] In fact, as shown in Graph 11, the increasing wage gap was more explained by decline of the relative wages on the bottom part of the distribution than what had happened on the top.



Source: Author' elaboration based on data from EPH (INDEC).

Furthermore, during the second half of the decade unemployment increased in all educational groups, even among the most educated. In particular, as shown in Table 2, unemployment rate of those that have not completed the secondary level was around 10% in average during the first subperiods and 20% in the second part of the nineties; for those graduates of the university level these figures were 4% and 7%, respectively.

On the other hand, the argument about sectoral changes applying Stolper-Samuelson theorem is based on evidence regarding the strong reduction of employment in manufacturing activities and the increase in professional services and the public sector, under the assumption that the Argentine manufacturing industry is, in relative terms, intensive in low-skilled labour. However, it does not seem to be completely adequate to apply this theorem for non-tradable sectors, like financial and public sectors, whose dynamism during the nineties was due to reasons totally different to economic openness. Also, as shown in Gasparini (2003), not only was employment in the low-tech manufacturing industry decreasing but a similar process was also verified among high-tech manufacturing, evidence that would not be completely consistent with that hypothesis.[29]

On the contrary, our argument emphasises that the fast reduction of the tariffs together with an appreciated exchange rate strongly affected international competitiveness of the manufacturing industry as a whole leading to a strong process of destruction of manufacturing enterprises. Additionally, the need to compete with imported industrial goods induced the acquisition of new technologies, a process that was favoured by the

[28] Marshall (2000) suggests that the process of decentralization of the bargaining of wages would be another factor associated to increasing salary dispersion and inequality. Additionally, from the comparative perspective with other countries, she concludes that the economic and labour market performance was more important than changes in labour legislations in the worsening in income distribution. However, these changes are not irrelevant but reinforced the negative effects of the macroeconomic regime.

[29] Additionally, studies for Argentina conclude that the most important distributive changes were due to the skilled-biased technological change while trade liberalization had a lower impact. In particular, Galiani and Sanguinetti (2003) and Cicowiez (2002) found a small increase in income inequality associated to trade openness and the reduction of tariffs.

change in relative prices. Therefore, besides employment reduction as a consequence of plants destruction, those enterprises that survived carried out a process of substitution of labour for capital resulting in a considerable reduction in the elasticity of labour demand. These productivity improvements were achieved both with a higher investment rate and with reorganization of the productive process (Altimir *et al.* (2002), Damill *et al.* (2002)).

Summing up, from the first perspectives the idea seems to rise that the distributive worsening during the nineties was the "necessary cost to assume" in order to achieve a more efficient and more competitive productive structure. As it is suggested, although the existence of an unequal effect associated to trade opening and integration can be accepted, there is evidence that these processes also allow higher economic growth rates to be reached, which, in turn, positively affect welfare through the reduction of poverty levels. However, that hypothesis was not completely verified in the case of Argentina during the nineties. Part of the high poverty levels was directly due to unsustainability of the macroeconomic regime. Towards the end of the regime, the strategy for sustaining the convertibility generated higher unemployment and poverty levels. Additionally, during the nineties inequality has increased together with growing unemployment rates. Therefore, the trade-off between inequality and unemployment implicit in the Unified Theory has not been verified.

4. Reversion and perdurability after the macroeconomic regime change[30]

As mentioned in Esquivel and Maurizio (2005), the worsening of living conditions, especially during the last years of the convertibility, has been of such a magnitude that its effect should be taken into account when evaluating the performance of the new macroeconomic regime.

During 2002 Argentina experienced an economic and social crisis of unusual magnitude as a consequence of the fall of the convertibility plan. But as the depth of the economic and social crisis does not have records in the country, the recovery that began towards the second part of the year was also intense, especially, with respect to the employment dynamic. In particular, the net generation of jobs –even excluding those originated in the "Plan Jefes y Jefas de Hogar Desocupados (PJJHD)"–[31] was higher than the one that could be predicted taking into account the GDP evolution.

Immediately after the exit of the convertibility the nominal exchange rate increased in a significant way. The over-adjustment that this variable experienced reflected the high degree of uncertainty in an economy that had suffered an abrupt break of the macroeconomic rules that had persisted during the previous decade. This sharp depreciation of the peso implied a strong increase of the domestic prices. However, the adjustment of prices turned out to be smaller in magnitude than that of the exchange rate being it the reason why the real exchange rate was duplicated towards June 2002.

[30] This section is partly based on Damill, Frenkel and Maurizio (2007).
[31] The PJJHD was implemented in 2002 by the National Government to response to the prevailing critical situation in the country. It established a fixed amount of $150 aimed at unemployed household heads with children up to 18 years old. In 2003 it reached about 2 million households to systematically reduce its coverage later.

Unlike previous experiences, this significant rise of the general prices level did not result in an inflationary process. That was due to the deep economic depression and the already very vulnerable labour and social situation before the regime change. In particular, the weakness of the domestic demand imposed a limit to the *pass through* from devaluation to consumer prices. Also, the high unemployment levels explained, in part, the absence of mechanisms of wage indexation, what made the increase in prices shock directly the purchasing power of the remunerations. The lack of liquidity caused by the maintenance of restrictions to the use of deposits in the banks was another reason that contributed to this dynamic of the domestic prices. All these elements stopped the propagation mechanisms of the characteristic inflationary impulses of previous devaluations, giving place to a novel situation in the recent Argentine economic history.

The domestic product experienced different phases after the change of macroeconomic regime. Initially, it continued falling during the first quarter of 2002, it was stabilized in the following quarter and it started increasing from the third one, when it began a process of sustained growth that is extended until the present time (Graph 5). In fact, by mid-2005 it had already exceeded the maximum level reached in 1998. The current economic regime leans, fundamentally, on the maintenance of a high real exchange rate. This constitutes a determinant factor of the strong recovery that aggregate activity level has registered since the middle of 2002 given that it has allowed raising the competitiveness of tradable sectors.

It is argumented –see for instance, Frenkel (2005)–, that a regime of high real exchange rate promotes the generation of new job positions through its lower proclivity to the macroeconomic unbalance (in comparison with a regime of low exchange rate), the change in the composition of the domestic production (more biased to tradable sectors) and the alteration of the relative prices favourable to a higher use of labour. However, agreeing or not with these argumentations, immediately after the crisis more short-term reasons seem to have prevailed in the explanation of the performance of the employment generation, such as the high degree of sub-utilization of installed capacity.[32]

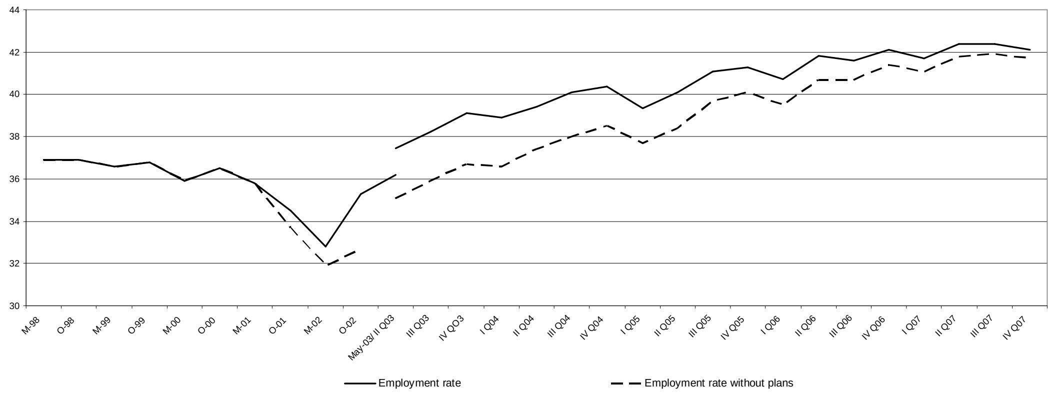
The change of macroeconomic model has had very positive effects on the dynamism of labour market and income generation.[33] After the negative shock caused by the collapse of the previous regime, all labour indicators have been able to break the trend towards the systematic worsening, although with different intensity. In particular, employment has experienced three clearly differentiated phases (Graph 12). The first of them covers the next immediate semester to the model's change between October 2001 and May 2002, which was characterized by a very important contraction of the aggregate employment level. In the second one, between May 2002 and the fourth quarter of the same year, the genuine employment was able to restrain its fall, while the implementation and expansion of the PJJHD implied the generation of a very significant amount of new job positions.

The third phase begins at the end of 2002 and lasts until the present time, period in which a consolidation and accelerated recovery of the employment took place. This process has been characterized by a high creation of new jobs by the private sector that

[32] Beccaria, Esquivel and Maurizio (2005).
[33] For an analysis of the labour market situation after the Convertibility also see Monza (2002).

more than compensated the reduction of beneficiaries of PJJHD that is being verified from about the middle of 2003.[34] In the fourth quarter of 2002, the employment rate (including this employment plan) was already higher than the one observed one year before (last observation of the convertibility period) while in the third quarter of 2003 it exceeded the level of 1998, the maximum of the second half of the nineties. On the other hand, the employment level excluding these plans was completely recovered from the post-devaluation fall in the second quarter of 2003.

Graph 12
Employment rate (with and without employment plans)
Total urban centres
May 1998-IV quarterly 2007



Source: Author' elaboration based on data from EPH (INDEC).

The dynamics of the aggregate employment has been particularly positive in the case of the industrial activities. From the fourth quarter of 2002 the manufacturing sector was finally able to break the falling trend in the employment level that it had been experiencing during last decade. In particular, it experienced a reduction during the first three quarters of 2002, to face then an increasing process that lasts until the present time. Since the fourth quarter of 2002 until the third quarter of 2007 it increased about 37% being, in that moment, 31% higher than the last register of the convertibility.

Although the industrial employment has had a very good performance, since 2003 the important growth of employment was sectorally generalized, as it can be seen in Table 3. In particular, it is observed that from that year on, commerce and construction, together with industrial activities and financial services have had the highest contribution to total employment generation. On the whole, they explain about 70% of the new net employment. The intensity of the occupational growth was related to the dynamic shown by the activity level, since it was stronger where the domestic product grew more (especially in manufacture, construction and commerce).

Another important dimension to characterize the evolution of the aggregate employment is the occupational category that allows distinguishing the workers as registered wage earners (registered in social security), non-registered wage earners, non-wage earners (own-account workers and employers) and unpaid family workers. Along the whole period (2003-2006), the registered job positions increased 30% explaining 72% of the total new employment generated. On the other hand, the non- registered jobs rose 15%

[34] The beneficiaries of the PJJHD that fulfilled the programme's counterpart work requirements represented around 8% of the total urban employment in 2003 whereas in 2006 their participation was reduced to 3%.

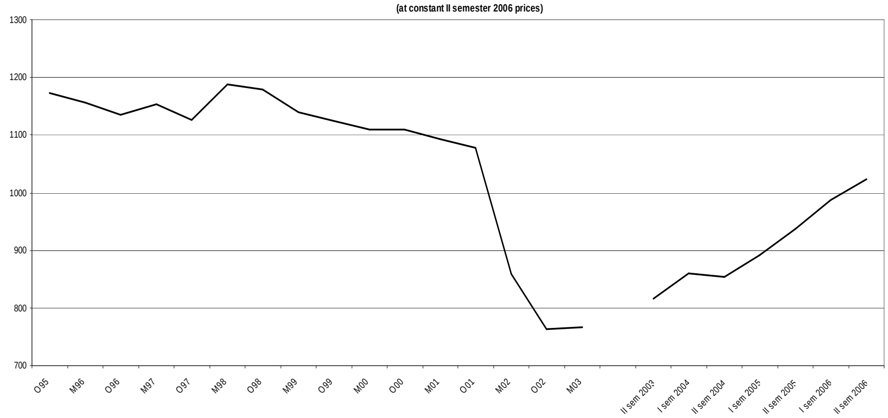
and they explain 29% of the employment creation. The independent activities have shown a lower dynamism given that during the period grew only 5% (Table 3).

Therefore, registered wage earners have shown a significant and sustained growth which has allowed them to raise their participation in the total employment from 40% in 2003 to 44% in 2006. However, after so many years of persistent unfavourable indicators, the generation of new jobs, especially at the beginning of the macroeconomic recovery, has been characterized by an important degree of labour precariousness. Indeed, in the first quarter of 2006 non-registered wage earners still represented about 42% of the total salaried employment (Table 3). Precariousness is one of the most important signs of the strong deterioration suffered by the labour market, especially after the second half of the past decade. Given that, as it will be seen later, the precarious workers obtain, on average, 40% of the remunerations of the registered workers, this aspect turns to be important to explain the existing inequality inside the workforce.

Lastly, the recovery of the occupational level was spread among workers with different skills, although with lower intensity among those with complete primary education or less (Table 3). This relatively homogeneous evolution in the occupations in terms of the educational level constitutes a novel fact given that previous experiences of employment recovery were biased almost exclusively towards the most skilled job positions. This dynamism is explained, partly, by the one shown at a sectoral level, especially with respect to the employment creation by the construction activities that, in relative terms, demand lower skilled workforce.

Similar to the evolution of employment, three phases can be identified in the evolution of real wages. There was a first phase after the devaluation in which real wages fell, a second phase of stabilization and a third phase of recovery and growth that started at the beginning of 2003. The increase in domestic prices after the devaluation had a direct negative effect on the purchasing power of wages. Between October 2001 and October 2002 real wages fell around 30%, although more than two thirds of the fall was verified in the first semester after the exit from the convertibility. After this strong reduction, since October 2002 nominal labour incomes started to grow at a similar pace than prices, hence the figure of May 2003 was similar to that of October 2002 (Graph 13). From 2003, both the greater dynamism of the labour demand and the price stability allowed an increase in the real labour incomes after the long period of continuous fall. However, given the strong previous reduction, at the end of 2006 real wages were, on average, lower than those at the end of the convertibility.

Graph 13
Real labour income. Excludes employment plans.
Total urban centres. October 1995- II semester 2006
(at constant II semester 2006 prices)



Source: Author' elaboration based on data from EPH (INDEC).

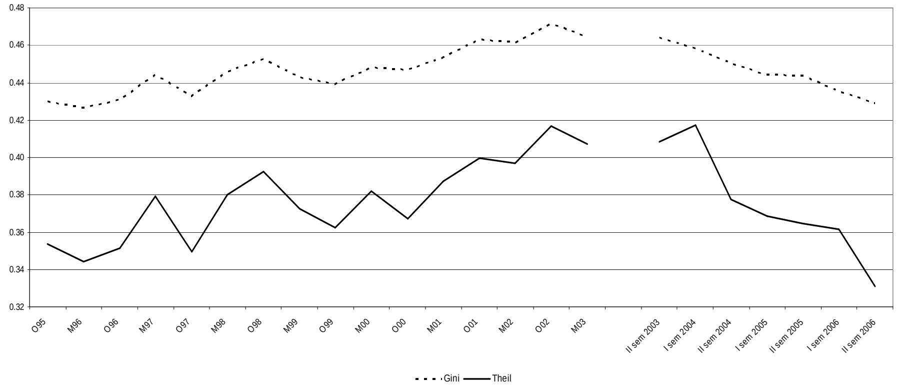
Real wages increased for all the groups of workers defined according to their occupational category, although with different intensities (Table 4). In particular, between 2003 and 2006 there was a 24% increase in real wages of registered wage earners whereas the increase for non-registered wage earners was lower but significant (17%). At the beginning of the new regime the increase of the labour income was even higher for the latter group than for the former. This fact constitutes a novel phenomenon since in the previous periods of wage recovery the wages had taken place fundamentally among registered wage earners.

The wages dynamic has also been different according to the educational level of workers. In particular, there has been a relative improvement in incomes of the less educated throughout this period: while the real income of workers with incomplete secondary school or less increased 30%, for those with complete university the raise was about 20% (Table 4). As a result, there has been a reduction in the wage gap between the extremes of the distribution that has contributed to the reduction of inequality among workers, reversing the previous increasing trend.

In particular, due to the generalized fall of real wages along the semester following the collapse of the convertibility, the Gini and Theil coefficients of labour income remained practically unchanged. After a small increase in inequality between May and October 2002 the trend towards less concentration of incomes started, in contrast with the process experienced throughout the nineties. The Gini coefficient of labour income fell from 0.464 to 0.429 between the second semester of 2003 and the second semester of 2006 (Graph 14 and Table 5).[35] The Theil coefficient verified a more intense reduction.

[35] This reduction is statistically significant with a 95% level of confidence.

Graph 14
Gini and Theil coefficiens of labour income
Excludes employment plans
Total urban centres. October 1995- II semester 2006



Source: Author' elaboration based on data from EPH (INDEC).

Therefore, the change of trend of inequality since 2003 implies the need to find factors associated to this dynamic. However, the estimates of the specific relative importance of each of them is hard because, on the one hand, since the macroeconomic regime change there have been several contemporaneous events that could have had an impact on income distribution. On the other hand, the proximity of the changes in macroeconomic and labour performance[36] do not allow us to differentiate completely the initial effect of the change in macroeconomic regime of the one associated to its "normal" functioning.

However, it is possible to mention at least some factors associated to the reduction of inequality. Regarding labour income distribution, the recovery in labour demand that allowed both growth in employment and wages across workers with different skills is likely to have been the base that made a reduction of the wages gap possible. However, probably, another factor that has contributed to this process has been the incomes policy implemented by the National Government since mid-2002 through lump-sum rises and increments in the minimum wage, measures that have greater impact on lower income groups, as it is the case of least skilled workers and non-registered wage earners.[37] Additionally, the process of registration of non-registered wage earners verified in the last years must also have triggered the growth in wages in the bottom extreme of the distribution.

Nevertheless, in spite of the reversal in the trend towards greater inequality, the concentration of income is still high due, partly, to the high level of income inequality prior to the change of regime (Table 6). In the second semester of 2006, whereas the first quintile of employed workers received 4% of total wages, the fifth quintile received approximately 47%. Moreover, average income of the latter group was 12 times the average income of the first quintile (this ratio was of 14 times at the end of 2003).

As mentioned, the occupational category is one of the dimensions through which the strong heterogeneity among workers is clearly seen, and in particular, the condition of being registered or not among the wage earners. In effect, in the second semester of 2006, those wage earners not covered by social security received, on average, incomes

[36] Especially because, due to the already mentioned reason, the analysis ends in 2006.
[37] Even though the non-registered wage earners are not covered by labour legislation, it is argued that the wages earned by those workers that are covered by social security have a certain impact on the wages paid to the rest of wage earners.

that represented 43% of the incomes of registered wage earners (Table 4). This fact reflects that despite the strong jobs creation, there still exist two clearly differentiated groups among workers: on the one hand, those workers who receive high incomes and are covered by social security; on the other hand, a group of workers with low wages, precarious jobs and without social security protection.

As a result of the favourable evolution of employment and the recovery of wages, as well as public policies, real HPCI started a process of growth since 2003. Between the second semester of that year and the end of 2006 they grew about 40% in real terms allowing the full recovery of purchasing power after the strong reduction in 2002 (Table 7).

Contrary to what happened among workers, in the semester that followed the exit from the convertibility family incomes inequality rose, mainly as a consequence of the increase in unemployment. This is reflected in the deterioration of the Gini coefficient of HPCI that passed from 0.520 to 0.541 between October 2001 and May 2002 (Table 7). After the maximum concentration level reached in May 2002, the concentration trend reversed. One of the factors that initially explained this turning point was the implementation of the PJJHD since, as mentioned, it gave employment and/or incomes to the poorest families. Its impact is reflected in the fall of the Gini coefficient of total family income between May and October 2002, which in absence of this programme would have recorded 0.505 instead of 0.489. As from 2003 the inequality among households has shown a sustained decreasing trend that allowed a 5 p.p reduction in the Gini coefficient, from 0.532 to 0.483, between the second half of that year and the second half of 2006.

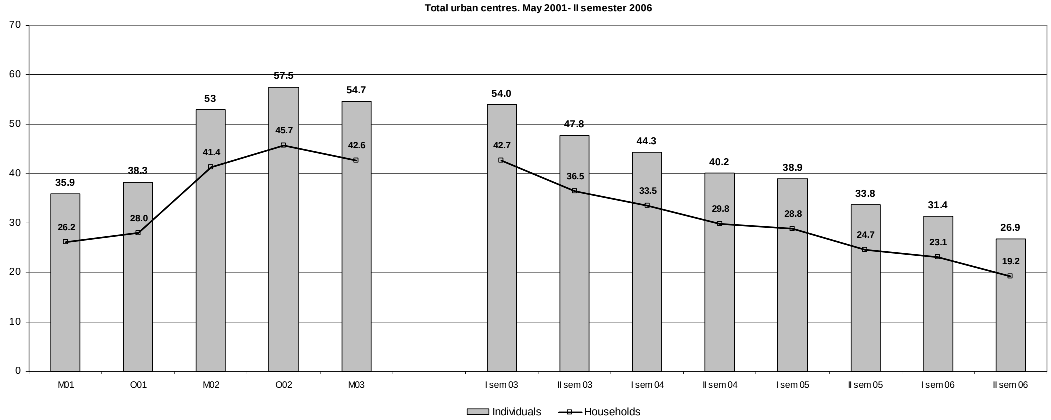
Additionally to the factors associated to the reduction of the wage gap, the significant fall of open unemployment (which affects to a greater extent those households in the lower extreme of the distribution) due to the increase of employment and several public policies implemented during the last years should be factors associated to the decline of household inequality. Among the latter, the most important were the implementation and extension of the PJJHD together with the recovery of the value of real pensions and the extension in their coverage.

In spite of this dynamic, the concentration of family income is still extremely high. In the second half of 2006, the 20% poorest households received only 4% of total income, whereas the fifth quintile captured 50% (Table 8). This situation of high inequality avoids farther poverty reduction as a result of economic growth, although, as will be shown below, the fall of poverty rates has been important.

Indeed, after reaching 57,5% of poor people in October 2002 in the total urban centres, poverty has been reducing systematically since 2002. This allowed the percentage of individuals in poor households to decrease 27 p.p. –from 54% to 26.9%- between the first semester of 2003 and the second semester of 2006 (Graph 15), while in the case of indigent individuals the proportion fell 19 p.p. in the same period –from 27,7% to 8,7%.

Given the strong dynamic of the poverty rate during the last years it is important to quantify the relevance of the changes in average real wages and in income distribution on social conditions. In particular, the variations in the level of poverty can be decomposed in two effects[38]: on the one hand, the change experienced as a consequence of the variations in the household average real income, maintaining the distribution constant –growth effect- and, on the other hand, as a consequence of distributive effects, with a constant average income –distribution effect. The growth effect can in turn be decomposed in the "inflation effect" and in the "nominal income effect". The former effect indicates how large the variation in the poverty level would have been with constant nominal incomes and distribution. The latter quantifies the impact of the changes in incomes under the assumption that prices and distribution remain constant. The results are shown in Table 9.[39]

Graph 15
Poverty rate
Total urban centres. May 2001- II semester 2006



Source: Author' elaboration based on data from EPH (INDEC).

During 2001 the fall in household total incomes explained 75% of the increase in poverty (income effect), even though deflation slightly attenuated the fall in real incomes since it made the poverty line lower. On the other hand, the deterioration in income distribution explained the other 25% of the increase in poverty. As from that moment, the distribution effect loses its relevance and the increases in poverty levels become mainly explained by the deterioration of real incomes due to inflation in the first semester of 2002. In particular, between May and October 2002, the increases in family incomes (the negative sign of the nominal income effect) were not sufficient to compensate for the price increases. Therefore poverty continued to rise, although at a lower pace than in the previous semester. The increase in family incomes in this period is explained, to a great extent, by the rapid expansion of the PJJHD.[40]

The negative trend of the social situation reversed since 2003. As it was already mentioned, the joint result of the increase in employment and wages led to a process of growth in family incomes and a simultaneous gradual improvement in their distribution. As it is shown in Table 9, both the "growth effect" and the "distribution effect" have been important in the reduction of poverty (although the effect of the former was clearly greater, of 79% against 21% of the latter). In addition, real incomes were able to rise

[38] This decomposition is based on Mahmoudi (1998).

[39] The negative sign means that the effect in consideration worked in the opposite direction than the variation of poverty.

[40] Even though the PJJHD was correctly focused on the poorest population, the impact on the poverty incidence was small because the amount of the transfer was low in relation to the value of the poverty line. There has been a larger effect in the case of indigence. In addition, currently the impact of the plan on both indicators is small given the reduction in the number of beneficiaries since 2003.

despite the increase in the value of the poverty line (which is reflected in the negative sign of the inflation effect), especially in the more recent periods.

Therefore, the improvement of labour market conditions appears as a very important factor that has contributed to the strong poverty reduction in the country. However, in spite of this favourable dynamic in social conditions, deprivation levels still are very high where one of three persons continues living with incomes below the poverty line.

Additionally, poverty incidence and evolution have not been homogeneous across households with different composition and household head labour status. On the contrary, as shown in Maurizio *et al.* (2008) while in the second semester of 2006 households with children (18 years old or less) represented near to 49% of total households, this figure reached 76% among poor families. This fact clearly indicates the significant differences between the poverty rates specific of each both groups of households: 30% in the case of households with children and 9% among the rest of families. Furthermore, the poverty situation is worse among households with children but without spouse, especially in the case of female household head. Therefore, children constitute a group with very high social vulnerability. In particular, while the poverty rate among individuals in the second semester of 2006 was 27%, this figure raised to 40% among children. Furthermore, the reduction of poverty since 2006 was stronger among households without children indicating that the rest of the households benefited with less intensity with the improvement of social conditions registered during last years.

In relation to the labour status of the household head, as expected, poverty incidence is higher among unemployed and inactive (without any pension) heads of family. However, it is worth pointing out that to get a job is not enough to exit from poverty. At the end of 2006 approximately 33% of households whose head was employed as non-registered wage earners and 27% working as independent worker was poor. On the contrary, only 7% of the household heads working in a registered job was poor. In fact, 70% of poor households' heads are employed.

This panorama indicates that not only the lack of a job is a factor associated to the still high poverty rates in the country but also the deficient occupational insertion characterized by labour precariousness and part-time work. This constitutes the *poor worker phenomenon.*

Therefore, additionally to the macroeconomic regime that enables a strong and sustainable job generation, a labour policy oriented to improve the quality of jobs as well as to reinforce the real wage recovery is needed. However, this will probably not be enough to significantly reduce poverty incidence in the short and medium term. It is necessary, then, to complement labour and income policies with social policies focused on riskier groups –households with children, adults without pensions and unemployed people– in order to reach a reasonable level of social welfare.

5. CONCLUSIONS

Along the last three decades Argentina experienced very dramatic changes in its social composition, labour structure and income distribution as a result of the macroeconomic performance, structural reforms and changes in the productive and labour structure. From middle-70, the labour difficulties, the fall of real wages and the increase of inequality have been eroding the basic principle of social cohesion, deepening the distance of incomes among social strata. Being a country characterized in the region by a very low level of inequality, widespread labour protection and reduced poverty incidence, the country has been experiencing a systematic worsening of its social conditions.

From the previous analysis it is possible to conclude, on the one hand, that the macroeconomic regime matters in terms of distributional and living conditions outcomes. In particular, Argentina shows that it is possible to experience very high GDP growth rates together with increasing and elevated unemployment rates and precariousness and inequality. During the nineties, the combination of very quick trade opening, real appreciation of the currency and structural reforms implied a very weak labour demand even under a context of significant dynamic of the GDP.

On the other hand, the negative effects that the macroeconomic configuration has on the labour market and the income distribution persist even after the economy returns to its growth path. In this sense, it seems that a pattern has emerged where the successive crises acted worsening the income distribution as long as the recovery cycles founded limits for the complete reversion of these trends.

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ANNEX

Table 1
Mincer Equations
Hourly wages. Wage earners.
Greater Buenos Aires

Covariates	1991	1993	1998	2001
Educational level				
Incomplete primary or less	-0.091	-0.111	-0.111	-0.186
	(-2.23)*	(-2.89)**	(-2.55)*	(-3.43)**
Incomplete secondary	0.235	0.144	0.181	0.167
	(7.01)**	(4.75)**	(5.94)**	(4.35)**
Complete secondary	0.473	0.384	0.474	0.418
	(12.56)**	(11.02)**	(13.52)**	(9.64)**
Incomplete university	0.566	0.565	0.756	0.64
	(11.54)**	(12.74)**	(17.10)**	(11.68)**
Complete university	0.872	0.892	1.065	0.957
	(17.29)**	(19.07)**	(22.64)**	(15.72)**
Other personal characteristics				
Men	0.084	0.204	0.209	0.105
	(1.93)	(4.79)**	(5.26)**	(2.34)*
Household' head	0.121	0.142	0.119	0.159
	(3.09)**	(4.09)**	(3.52)**	(3.70)**
Married	0.106	0.015	0.055	0.09
	(3.46)**	(0.5)	(2.10)*	(3.03)**
Age	0.04	0.076	0.057	0.062
	(4.05)**	(6.87)**	(4.97)**	(3.79)**
Age*Age	0.00	-0.001	-0.001	-0.001
	(3.50)**	(-6.46)**	(-4.24)**	(-3.30)**
Industry				
Construction	0.149	0.102	0.076	0.26
	(2.58)*	(1.87)	(1.6)	(3.88)**
Trade	-0.022	-0.069	-0.136	-0.11
	(-0.61)	(-2.10)*	(-4.01)**	(-2.84)**
Transport	0.015	-0.033	-0.112	-0.086
	(0.32)	(-0.82)	(-2.84)**	(-1.9)
Financial services	0.113	0.072	-0.011	0.138
	(2.40)*	(1.67)	(-0.29)	(2.98)**
Personal services	-0.218	-0.207	-0.09	0.001
	(-3.89)**	(-4.45)**	(-2.04)*	(0.02)
Domestic services	0.096	0.105	0.234	0.229
	(1.96)*	(2.13)*	(4.60)**	(4.09)**
Public sector	-0.224	-0.212	-0.117	-0.074
	(-5.24)**	(-5.47)**	(-3.11)**	(-1.65)
Other industries	0.085	0.013	0.059	0.042
	(1.78)	(0.29)	(1.43)	(0.9)
Occupational category				
Non-registered wage earner	-0.163	-0.084	-0.188	-0.187
	(-5.44)**	(-3.14)**	(-7.43)**	(-6.08)**
Size of the firm				
16-50 workers	0.007	0.018	0.129	0.127
	(0.21)	(0.6)	(4.36)**	(3.68)**
51-100 workers	0.062	0.065	0.124	0.189
	(1.38)	(1.66)	(3.42)**	(4.16)**
More than 100 workers	0.163	0.156	0.277	0.359
	(4.68)**	(5.01)**	(9.27)**	(10.05)**
Underemployed	0.31	0.307	0.231	0.357
	(9.10)**	(10.38)**	(8.67)**	(12.03)**
Lambda	0.161	0.327	0.206	0.155
	(2.13)*	(4.01)**	(2.31)*	(1.23)
Constant	8.516	-1.036	-0.742	-0.935
	(34.88)**	(-3.89)**	(-2.55)*	(-2.18)*
Observations	2042	2494	2892	2440
R-squared	0.33	0.31	0.43	0.45

Absolute value of t statistics in parentheses
* significant at 5%; ** significant at 1%

Source: Author' elaboration based on data from EPH (INDEC).

Table 2
Unemployment rate by educational level
Average in each subperiods

Educational level	May 1991- May 1995	May 1995- October 1998	October 1998- October 2001
Incomplete primary or less	11%	21%	21%
Complete primary	11%	19%	19%
Incomplete secondary	12%	20%	20%
Complete secondary	9%	16%	17%
Incomplete university	9%	15%	17%
Complete university	4%	7%	7%
Total	10%	17%	17%

Source: Author' elaboration based on data from EPH (INDEC).

Table 3
Evolution of employment by category, educational level and sector
Total urban centres. II semester 2003- II semester 2006
Excludes employment plans

	II Sem. 2003	I Sem. 2004	II Sem. 2004	I Sem. 2005	II Sem. 2005	I Sem. 2006	II Sem. 2006	Var.2003/ 2006	Contrib. to employment growth
Total employment	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**16%**	**100%**
Category									
Registered wage earner	39.6	40.8	40.5	41.2	41.9	43.6	44.0	29%	72%
Non-registered wage earner	31.5	32.0	32.1	32.2	31.7	31.2	31.2	15%	29%
Own-Account	22.4	21.5	21.6	21.4	20.9	20.1	19.4	1%	1%
Employer	4.0	4.0	4.4	4.0	4.3	4.1	4.2	21%	5%
Family worker without remuneration	2.5	1.7	1.4	1.1	1.3	1.1	1.1	-26%	-3%
% Non-registered / Total wage earners	44.3	44.0	44.2	43.9	43.1	41.7	41.4		
Educational level									
Incomplete primary or less	7.2	7.2	6.9	6.7	7.1	6.8	6.9	13%	6%
Complete primary	21.7	22.5	22.4	22.5	21.5	21.4	20.5	10%	13%
Incomplete secondary	18.3	17.4	17.3	17.0	17.0	16.9	17.0	8%	9%
Complete secondary	20.5	20.8	21.3	21.4	21.1	21.9	21.9	24%	31%
Incomplete terciary	13.6	14.0	13.9	14.0	14.1	14.2	14.5	23%	20%
Complete terciary	18.8	18.1	18.3	18.4	19.3	18.7	19.2	18%	22%
Sector									
Manufacture	14.4	14.6	14.9	14.9	14.4	14.7	14.2	15%	13%
Construction	7.4	8.0	8.1	8.2	8.8	8.6	8.9	39%	18%
Trade	24.7	25.0	25.4	24.0	24.3	23.9	24.4	15%	23%
Transport	7.0	6.8	7.3	7.0	7.0	6.7	6.6	8%	4%
Financial services	9.6	9.7	9.2	10.3	9.7	10.3	10.2	23%	14%
Personal services	6.7	6.4	6.4	6.3	6.5	6.5	6.7	15%	6%
Domestic services	8.1	7.9	7.9	8.1	8.0	8.0	8.1	17%	9%
Public sector	14.3	13.4	13.6	13.5	13.5	13.9	13.7	11%	10%
Other industries	7.2	7.8	7.0	7.5	7.5	7.2	6.9	12%	6%

Source: Author' elaboration based on data from EPH (INDEC).

Table 4
Evolution of real labour income by category and educational level
Total urban centres. II semester 2003- II semester 2006
Excludes employment plans

	II Sem. 2003	I Sem. 2004	II Sem. 2004	I Sem. 2005	II Sem. 2005	I Sem. 2006	II Sem. 2006	Var. 2003/2006
Total employment	816	860	854	892	938	987	1023	25%
Category								
Registered wage earner	1057	1104	1076	1123	1191	1244	1309	24%
Non-registered wage earner	480	496	518	529	541	551	559	17%
Own-Account	639	682	698	713	744	810	806	26%
Employer	2020	1877	1793	2139	2100	2174	2210	9%
% Non-registered / registered wage earners	45%	45%	48%	47%	45%	44%	43%	
Educational level								
Incomplete secondary or less	543	586	582	617	627	691	704	30%
Complete secondary-incomplete terciary	847	886	891	915	973	1019	1045	23%
Complete terciary	1441	1533	1474	1565	1620	1669	1731	20%

Source: Author' elaboration based on data from EPH (INDEC).

Table 5
Real labour income
Average, Gini coefficient and Theil coefficient
Total urban centres. October 1995- II semester 2006
Excludes employment plans

	Average real labour income (at constant II Sem. 2006 prices)	**Gini coefficient**			**Theil coefficient**		
		Coefficient	Interval (95% of confidence)		Coefficient	Interval (95% of confidence)	
Oct-95	1172	0.4301	0.4227	0.4388	0.3535	0.3362	0.3774
May-96	1157	0.4266	0.4173	0.4346	0.3442	0.3247	0.3630
Oct-96	1136	0.4308	0.4211	0.4395	0.3513	0.3316	0.3704
May-97	1153	0.4442	0.4357	0.4552	0.3793	0.3572	0.4065
Oct-97	1126	0.4328	0.4244	0.4412	0.3498	0.3316	0.3640
May-98	1188	0.4458	0.4361	0.4562	0.3800	0.3536	0.4053
Oct-98	1180	0.4527	0.4420	0.4646	0.3926	0.3652	0.4206
May-99	1140	0.4428	0.4321	0.4516	0.3727	0.3513	0.3946
Oct-99	1125	0.4393	0.4323	0.4494	0.3623	0.3437	0.3864
May-00	1109	0.4482	0.4377	0.4570	0.3821	0.3574	0.4045
Oct-00	1109	0.4471	0.4396	0.4546	0.3673	0.3535	0.3825
May-01	1092	0.4535	0.4453	0.4636	0.3870	0.3687	0.4123
Oct-01	1078	0.4634	0.4513	0.4715	0.3995	0.3727	0.4212
May-02	859	0.4612	0.4524	0.4701	0.3971	0.3706	0.4226
Oct-02	764	0.4717	0.4573	0.4845	0.4170	0.3855	0.4488
May-03	766	0.4647	0.4529	0.4770	0.4072	0.3747	0.4404
II sem 2003	816	0.4642	0.4541	0.4763	0.4086	0.3806	0.4456
I sem 2004	860	0.4584	0.4477	0.4742	0.4173	0.3703	0.4945
II sem 2004	854	0.4504	0.4427	0.4620	0.3776	0.3529	0.4148
I sem 2005	892	0.4443	0.4369	0.4524	0.3685	0.3515	0.3881
II sem 2005	938	0.4438	0.4360	0.4499	0.3646	0.3444	0.3842
I sem 2006	987	0.4356	0.4270	0.4442	0.3615	0.3315	0.3959
II sem 2006	1023	0.4292	0.4232	0.4357	0.3310	0.3204	0.3445

Source: Author' elaboration based on data from EPH (INDEC).

Table 6
Quintile distribution of labour income
Total urban centres. II semester 2003- II semester 2006
Excludes employment plans

QUINTILE	II SEM 2003	I SEM 2004	II SEM 2004	I SEM 2005	II SEM 2005	I SEM 2006	II SEM 2006
1	3.7%	3.9%	3.9%	4.1%	4.0%	4.1%	4.0%
2	9.4%	9.6%	9.7%	9.8%	9.7%	10.1%	10.1%
3	14.3%	14.7%	14.9%	15.3%	15.5%	15.5%	16.1%
4	21.5%	21.3%	22.4%	22.0%	22.4%	22.1%	22.6%
5	51.0%	50.5%	49.1%	48.7%	48.4%	48.2%	47.2%
TOTAL	100%	100%	100%	100%	100%	100%	100%
5 quintile/ 1quintile	13.7	13.0	12.9	12.0	12.4	11.9	12.0

Source: Author' elaboration based on data from EPH (INDEC).

Table 7
Household per capita income
Average, Gini coefficient and Theil coefficient
Total urban centres. October 1995- II semester 2006
Includes employment plans

	Average real family income (at constant II Sem. 2006 prices)	Gini coefficient			Theil coefficient		
		Coefficient	Interval (95% of confidence)		Coefficient	Interval (95% of confidence)	
Oct-95	620.2	0.4880	0.4781	0.4987	0.4596	0.4287	0.5032
May-96	608.0	0.4859	0.4754	0.4979	0.4557	0.4282	0.4970
Oct-96	612.3	0.4972	0.4840	0.5131	0.4869	0.4469	0.5408
May-97	628.6	0.4917	0.4793	0.5033	0.4675	0.4340	0.5078
Oct-97	647.2	0.4957	0.4818	0.5184	0.4880	0.4292	0.6191
May-98	677.4	0.4990	0.4884	0.5087	0.4752	0.4486	0.5045
Oct-98	676.2	0.5049	0.4951	0.5160	0.4899	0.4558	0.5352
May-99	658.5	0.5015	0.4910	0.5136	0.4787	0.4510	0.5133
Oct-99	639.2	0.4928	0.4844	0.5030	0.4561	0.4334	0.4855
May-00	620.2	0.5035	0.4928	0.5136	0.4817	0.4580	0.5136
Oct-00	639.8	0.5098	0.4990	0.5184	0.4985	0.4643	0.5319
May-01	615.8	0.5158	0.5059	0.5278	0.5102	0.4795	0.5517
Oct-01	592.1	0.5206	0.5101	0.5301	0.5127	0.4789	0.5424
May-02	452.3	0.5416	0.5299	0.5526	0.5839	0.5434	0.6373
Oct-02	410.3	0.5203	0.5055	0.5320	0.5062	0.4733	0.5431
May-03	431.8	0.5192	0.5016	0.5332	0.5059	0.4613	0.5457
II sem 2003	510.5	0.5326	0.5228	0.5466	0.5611	0.5182	0.6107
I sem 2004	529.6	0.5212	0.5024	0.5478	0.5784	0.4797	0.7928
II sem 2004	561.4	0.5090	0.4931	0.5236	0.5181	0.4539	0.5874
I sem 2005	566.5	0.5004	0.4914	0.5098	0.4730	0.4458	0.5000
II sem 2005	622.0	0.4920	0.4833	0.5018	0.4489	0.4256	0.4765
I sem 2006	635.5	0.4861	0.4773	0.4965	0.4453	0.4239	0.4735
II sem 2006	695.7	0.4827	0.4733	0.4914	0.4421	0.4034	0.4925

Source: Author' elaboration based on data from EPH (INDEC).

Table 8
Quintile distribution of total household income
Total urban centres. II semester 2003- II semester 2006
Includes employment plans

QUINTILE	II SEM 2003	I SEM 2004	II SEM 2004	I SEM 2005	II SEM 2005	I SEM 2006	II SEM 2006
1	3.4%	3.9%	3.9%	4.0%	4.2%	4.1%	4.2%
2	8.3%	8.9%	8.9%	9.1%	9.0%	9.0%	9.2%
3	13.6%	13.7%	14.1%	14.1%	14.4%	14.3%	14.4%
4	21.6%	21.8%	22.1%	22.2%	22.2%	22.5%	22.6%
5	53.1%	51.7%	51.0%	50.6%	50.2%	50.0%	49.5%
TOTAL	100%	100%	100%	100%	100%	100%	100%

Source: Author' elaboration based on data from EPH (INDEC).

Table 9
Decomposition of poverty variation (household)
Total urban centres. October 2000- II semester 2006

Period	Variation (p.p)	**Growth effect**	Nominal income effect	Inflation effect	Residual	**Distribution effect**
Oct00-Oct01	4.3	75%	81%	-6%	0%	25%
Oct01-May02	13.4	93%	27%	65%	1%	7%
May02-Oct02	4.3	94%	-60%	160%	-6%	6%
Oct02-May03	-3.1	73%	53%	18%	1%	28%
II sem03-II sem04	-6.7	78%	98%	-21%	0%	22%
I sem04-I sem05	-4.7	75%	113%	-45%	8%	25%
II sem04-II sem05	-5.1	78%	140%	-63%	1%	22%
I sem05-I sem06	-5.7	72%	146%	-67%	-7%	28%
II sem05-II sem06	-5.5	78%	120%	-39%	-3%	22%
II sem03-II sem06	**-17.3**	**79%**	**118%**	**-37%**	**-1%**	**21%**

Source: Author' elaboration based on data from EPH (INDEC).